



Chapter 19

Network Layer: Logical Addressing

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19-1 IPv4 ADDRESSES

*An **IPv4 address** is a **32-bit** address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.*

Topics discussed in this section:

Address Space

Notations

Classful Addressing

Classless Addressing

Network Address Translation (NAT)

IPv4	IPv6
<u>IPv4 addresses</u> are 32 bit length.	<u>IPv6 addresses</u> are 128 bit length.
<u>IPv4 addresses</u> are <u>binary numbers</u> represented in decimals.	<u>IPv6 addresses</u> are <u>binary numbers</u> represented in <u>hexadecimals</u> .
<u>IPSec</u> support is only optional.	Inbuilt <u>IPSec</u> support.
<u>Fragmentation</u> is done by sender and forwarding routers.	<u>Fragmentation</u> is done only by sender.
No packet flow identification.	Packet flow identification is available within the <u>IPv6 header</u> using the <u>Flow Label</u> field.
<u>Checksum field</u> is available in <u>IPv4 header</u>	No checksum field in <u>IPv6 header</u> .
<u>Options fields</u> are available in <u>IPv4 header</u> .	No option fields, but <u>IPv6 Extension headers</u> are available.
<u>Address Resolution Protocol (ARP)</u> is available to map <u>IPv4 addresses</u> to <u>MAC addresses</u> .	<u>Address Resolution Protocol (ARP)</u> is replaced with a function of <u>Neighbor Discovery Protocol (NDP)</u> .
Internet Group Management Protocol (IGMP) is used to manage multicast group membership.	IGMP is replaced with Multicast Listener Discovery (MLD) messages.
<u>Broadcast messages</u> are available.	<u>Broadcast messages</u> are not available. Instead a link-local scope "All nodes" <u>multicast IPv6 address</u> (FF02::1) is used for broadcast similar functionality.
Manual configuration (Static) of <u>IPv4 addresses</u> or DHCP (Dynamic configuration) is required to configure <u>IPv4 addresses</u> .	Auto-configuration of addresses is available.

- **IPv4 Advantages**

- Connectionless Protocol and Best effort based.
- Addresses are easier to remember.
- Existing networks are already using it.
- Classful and classless addressing.
- Millions of addresses are wasted.
- Planning for excessive growth was not foreseen; addresses are running out.

- **IPv6 Advantages:**

- No more NAT (Network Address Translation)
- Auto-configuration
- No more private address collisions
- Better multicast routing
- Simpler header format
- Simplified, more efficient routing
- Built-in authentication and privacy support
- Flexible options and extensions
- Easier administration (say good-bye to DHCP)
- Large address space.
- Enhanced QoS.
- Efficient routing
- Built in security.
- Larger addresses harder to remember.
- Transition takes time and is not always smooth.
- Not always usable some machines have to be replaced.

Why IPv6?	IPv4	IPv6
IPv6 has more addresses	4.3 billion addresses	340 trillion trillion trillion addresses
IPv6 networks are easier and cheaper to manage	Networks must be configured manually or with DHCP. IPv4 has had many overlays to handle Internet growth, which demand increasing maintenance efforts.	IPv6 networks provide autoconfiguration capabilities. They are simpler, flatter and more manageable for large installations.
IPv6 restores end-to-end transparency	Widespread use of NAT devices means that a single NAT address can mask thousands of non-routable addresses, making end-to-end integrity unachievable.	Direct addressing is possible due to vast address space – the need for network address translation devices is effectively eliminated.
IPv6 has improved security features	Security is dependent on applications – IPv4 was not designed with security in mind.	IPSEC is built into the IPv6 protocol, usable with a suitable key infrastructure.
IPv6 has improved mobility capabilities	Relatively constrained network topologies restrict mobility and interoperability capabilities in the IPv4 Internet.	IPv6 provides interoperability and mobility capabilities which are already widely embedded in network devices.
IPv6 encourages innovation	IPv4 was designed as a transport and communications medium, and increasingly any work on IPv4 is to find ways around the constraints.	Given the numbers of addresses, scalability and flexibility of IPv6, its potential for triggering innovation and assisting collaboration is unbounded.

19-1 IPv4 ADDRESSES

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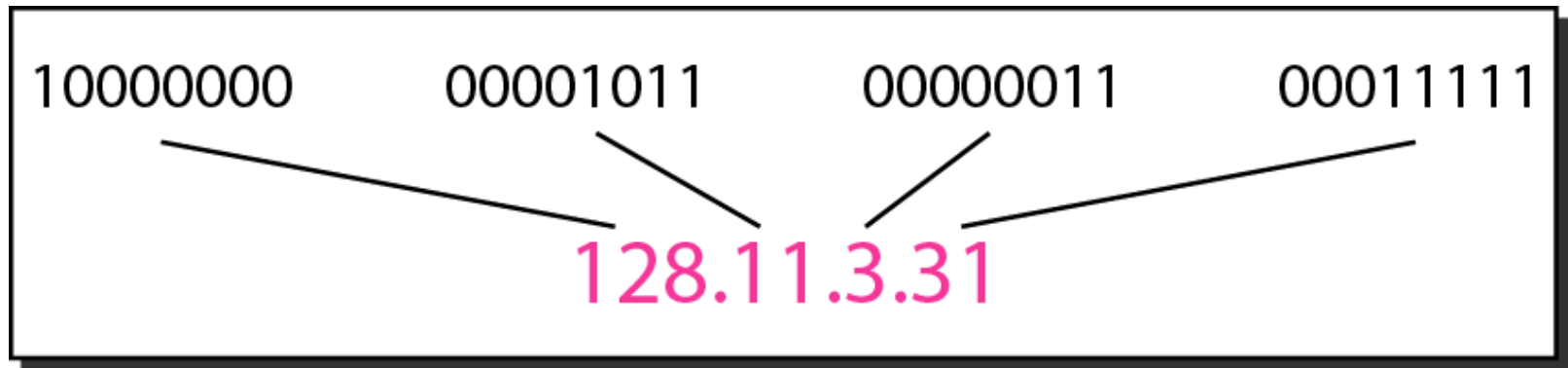
Note

An IPv4 address is 32 bits long.

**The IPv4 addresses are unique
and universal.**

**The address space of IPv4 is
 2^{32} or 4,294,967,296.**

Figure 19.1 *Dotted-decimal notation and binary notation for an IPv4 address*





Example 19.1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255



Example 19.2

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent (see Appendix B).

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010



Example 19.3

Find the error, if any, in the following IPv4 addresses.

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Solution

- a. There must be no leading zero (045).***
- b. There can be no more than four numbers.***
- c. Each number needs to be less than or equal to 255.***
- d. A mixture of binary notation and dotted-decimal notation is not allowed.***

Note

**In classful addressing, the address space is divided into five classes:
A, B, C, D, and E.**

Figure 19.2 *Finding the classes in binary and dotted-decimal notation*

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-decimal notation

Class	Address range
Class A	1.0.0.1 - 126.255.255.254
Class B	128.1.0.1 - 191.255.255.254
Class C	192.0.1.1 - 223.255.254.254
Class D	224.0.0.0 - 239.255.255.255
Class E	240.0.0.0 - 254.255.255.254

Figure 19.2 *Finding the classes in binary and dotted-decimal notation*

Five Different Classes of IPv4 Addresses

Class	First Octet decimal (range)	First Octet binary (range)	IP range	Subnet Mask	Hosts per Network ID	# of networks
Class A	0 — 127	0XXXXXXXX	0.0.0.0-127.255.255.255	255.0.0.0	$2^{24}-2$	2^7
Class B	128 — 191	10XXXXXXXX	128.0.0.0-191.255.255.255	255.255.0.0	$2^{16}-2$	2^{14}
Class C	192 — 223	110XXXXXX	192.0.0.0-223.255.255.255	255.255.255.0	2^8-2	2^{21}
Class D (Multicast)	224 — 239	1110XXXXX	224.0.0.0-239.255.255.255			
Class E (Experimental)	240 — 255	1111XXXXX	240.0.0.0-255.255.255.255			

Source:<https://medium.com/networks-security/tricks-to-remember-five-classes-of-ipv4-484c191678fb>

- Class A addresses were designed for large organizations with a large number of attached hosts or routers.
- Class B addresses were designed for midsize organizations with tens of thousands of attached hosts or routers.
- Class C addresses were designed for small organizations with a small number of attached hosts or routers.
- Subtract two(2) IP's is to reserve one IP for the network ID and one for the broadcast address.



Example 19.4

Find the class of each address.

- a.* 00000001 00001011 00001011 11101111
- b.* 11000001 10000011 00011011 11111111
- c.* 14.23.120.8
- d.* 252.5.15.111

Solution

- a.* The first bit is 0. This is a class A address.
- b.* The first 2 bits are 1; the third bit is 0. This is a class C address.
- c.* The first byte is 14; the class is A.
- d.* The first byte is 252; the class is E.

Table 19.1 *Number of blocks and block size in classful IPv4 addressing*

- One problem with classful addressing is that each class is divided into a fixed number of blocks with each block having a fixed size as shown in Table.

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Note

In classful addressing, a large part of the available addresses were wasted.

Netid and Hostid

- In classful addressing, an IP address in class A, B, or C is divided into netid and hostid. These parts are of varying lengths, depending on the class of the address.
- In class A, one byte defines the netid and three bytes define the hostid.
- In class B, two bytes define the netid and two bytes define the hostid.
- In class C, three bytes define the netid and one byte defines the hostid.
- Although the length of the netid and hostid (in bits) is predetermined in classful addressing, we can also use a mask(/32-bit number(max))
- The mask can help us to find the netid and the hostid.
- For example, the mask for a class A address has eight 1s, which means the first 8 bits of any address in class A define the netid; the next 24 bits define the hostid.
- This notation is also called **slash notation or Classless Interdomain Routing (CIDR) notation**. The notation is used in classless addressing.

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

Table 19.2 *Default masks for classful addressing*

Note

Classful addressing, which is almost obsolete, is replaced with classless addressing.

MASK

- A better way to define a block of addresses is to select any address in the block and the mask.
- A mask is a 32-bit number in which the n leftmost bits are **1s** and the **32 - n** rightmost bits are **0s**.
- In classless addressing the mask for a block can take any value from 0 to 32. It is very convenient to give just the value of n preceded by a slash (CIDR notation).
- The address and the **/ n** notation completely define the whole block (the first address, the last address, and the number of addresses).
- The first address in the block can be found by setting the **32 - n** right- most bits in the binary notation of the address to **0s**.
- The last address in the block can be found by setting the **32 - n** right- most bits in the binary notation of the address to **1s**.
- In IPv4 addressing, a block of addresses can be defined as

$x.y.z.t /n$

in which **$x.y.z.t$** defines one of the addresses and the **/ n** defines the mask.



Example 19.6

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32.

This is actually the block shown in Figure 19.3.



Example 19.7

Find the last address for the block in 205.16.37.39/28

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47

This is actually the block shown in Figure 19.3.

Note

**The number of addresses in the block
can be found by using the formula**

$$2^{32-n}.$$

Where n is the number of mask



Example 19.8

Find the number of addresses in Example 19.6.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} or 16.



Example 19.9

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number. This is particularly useful when we are writing a program to find these pieces of information. In Example 19.5 the /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Find

- a. The first address***
- b. The last address***
- c. The number of addresses.***

Example 19.9 (continued)

Solution

- a.** *The first address can be found by ANDing the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise.*

	205.	16.	37.	39
Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000
	205.	16.	37.	32

Example 19.9 (continued)

- b.** The last address can be found by ORing the given addresses with the complement of the mask. ORing here is done bit by bit. The result of ORing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.

	205.	16.	37.	39
Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111
	205.	16.	37.	47

Note

The first address in a block is normally not assigned to any device; it is used as the network address that represents the organization to the rest of the world.

Note

**Each address in the block can be considered as a two-level hierarchical structure:
the leftmost n bits (prefix) define the network;
the rightmost $32 - n$ bits define the host.**

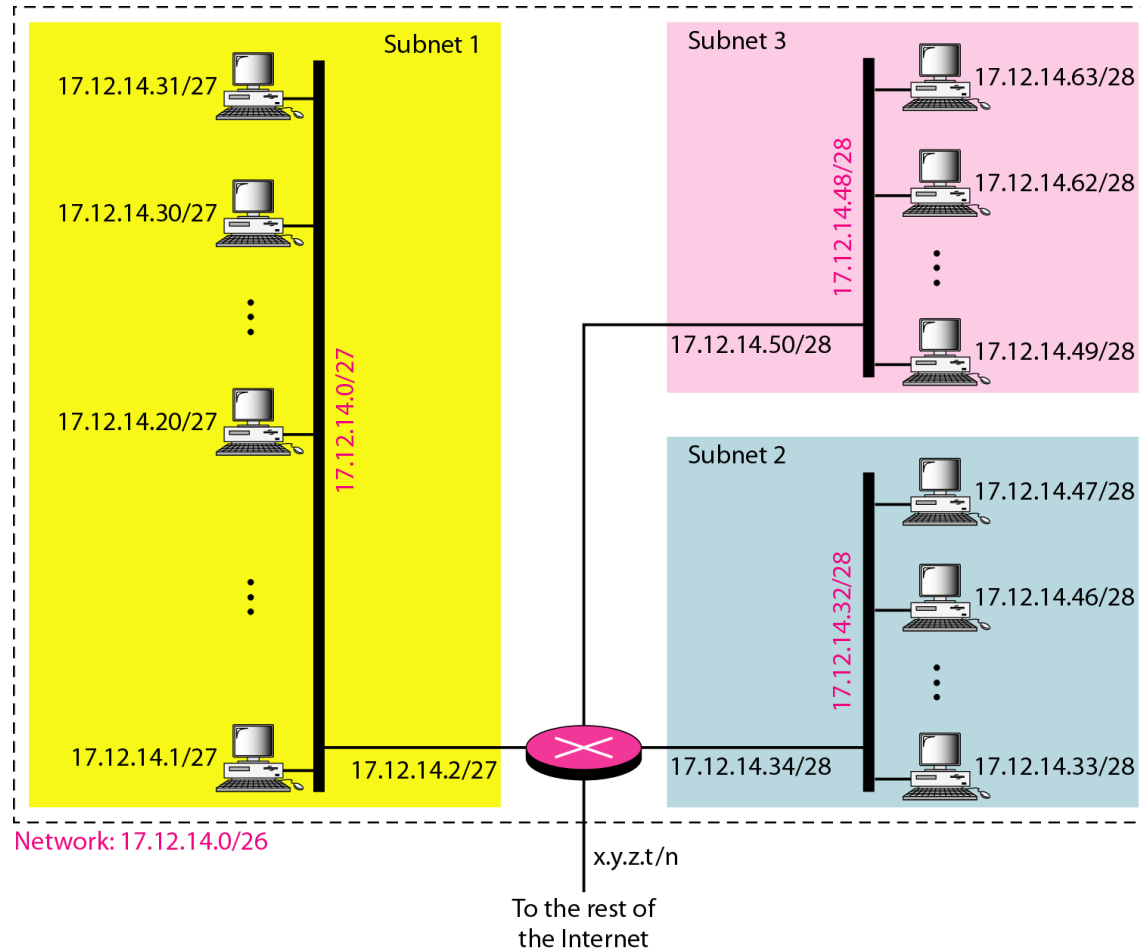
Configuration and addresses in a subnetted network

suppose an organization is given the block 17.12.14.0/26, which contains 64 addresses. The organization has three offices and needs to divide the addresses into three subblocks of 32, 16, and 16 addresses. We can find the new masks by using the following arguments:

1. Suppose the mask for the first subnet is n_1 , then 2^{32-n_1} must be 32, which means that $n_1 = 27$.
2. Suppose the mask for the second subnet is n_2 , then 2^{32-n_2} must be 16, which means that $n_2 = 28$.
3. Suppose the mask for the third subnet is n_3 , then 2^{32-n_3} must be 16, which means that $n_3 = 28$.

This means that we have the masks 27, 28, 28 with the organization mask being 26

Figure 19.7 *Configuration and addresses in a subnetted network*





Example 19.10

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- a. The first group has 64 customers; each needs 256 addresses.*
- b. The second group has 128 customers; each needs 128 addresses.*
- c. The third group has 128 customers; each needs 64 addresses.*

Design the subblocks and find out how many addresses are still available after these allocations.

Example 19.10 (continued)

Group 1

For this group, each customer needs 256 addresses. This means that 8 ($\log_2 256$) bits are needed to define each host. The prefix length is then $32 - 8 = 24$. The addresses are

$64 \text{ customer} \times 256 \text{ address} = 16384 \text{ address} = 2^{14}$

That means last 14(6+8) digits are set to 1 for last address with subtract 1 for broadcast address

<i>1st Customer:</i>	<i>190.100.0.0/24</i>	<i>190.100.0.255/24</i>
<i>2nd Customer:</i>	<i>190.100.1.0/24</i>	<i>190.100.1.255/24</i>
<i>...</i>		
<i>64th Customer:</i>	<i>190.100.63.0/24</i>	<i>190.100.63.255/24</i>
<i>Total = $64 \times 256 = 16,384$</i>		



Example 19.10 (continued)

Group 2

For this group, each customer needs 128 addresses. This means that 7 ($\log_2 128$) bits are needed to define each host. The prefix length is then $32 - 7 = 25$. The addresses are

<i>1st Customer:</i>	<i>190.100.64.0/25</i>	<i>190.100.64.127/25</i>
<i>2nd Customer:</i>	<i>190.100.64.128/25</i>	<i>190.100.64.255/25</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.127.128/25</i>	<i>190.100.127.255/25</i>
<i>Total = $128 \times 128 = 16,384$</i>		

Example 19.10 (continued)

Group 3

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed for each host. The prefix length is then $32 - 6 = 26$. 128 customer $\times$ 64 address = 8192 address = 2^{13} ($5+8$) in third and 4th octette. Add the 0.0.31.255 address to the 1st customer's last address to determine the 128 customers' last address. The addresses are

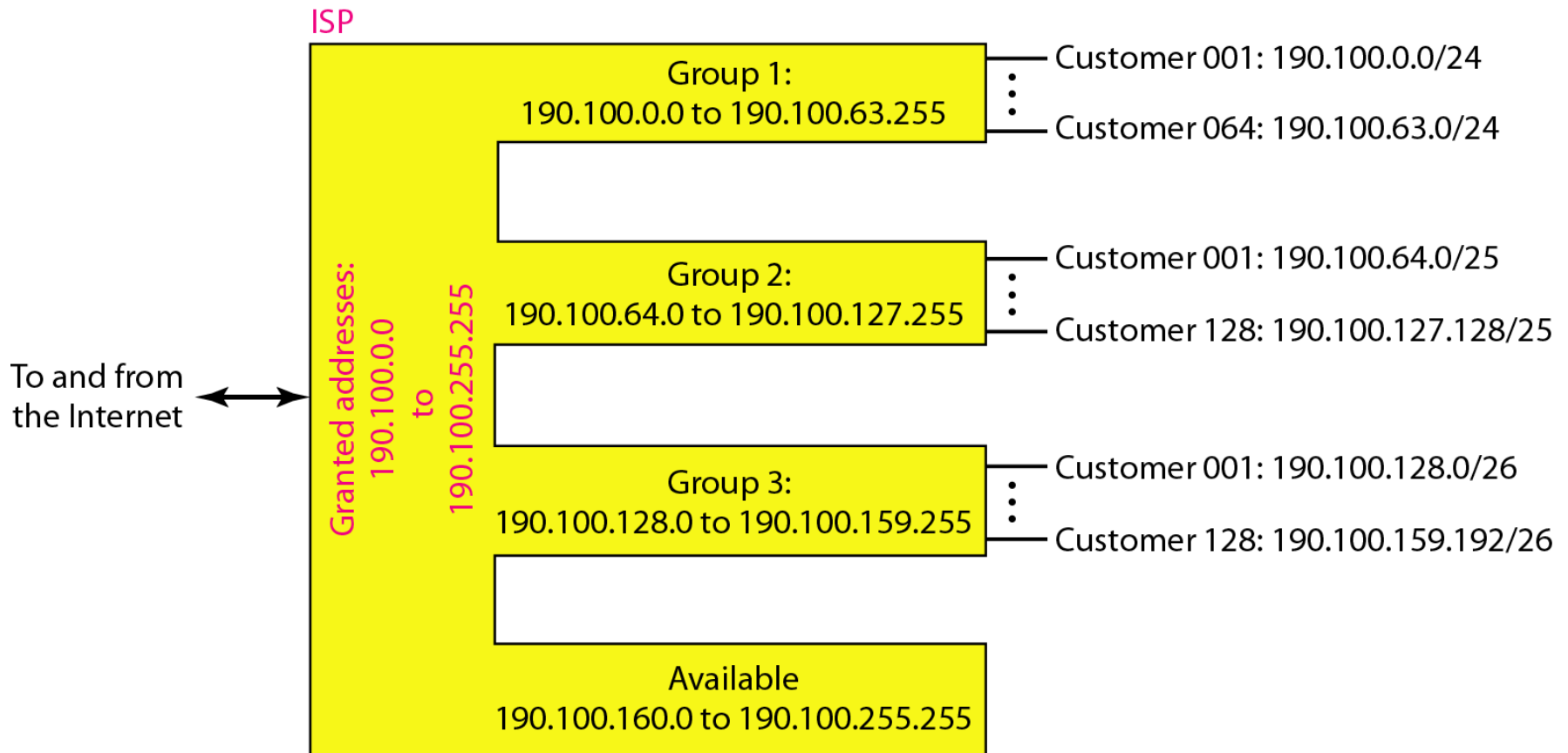
<i>1st Customer:</i>	<i>190.100.128.0/26</i>	<i>190.100.128.63/26</i>
<i>2nd Customer:</i>	<i>190.100.128.64/26</i>	<i>190.100.128.127/26</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.159.192/26</i>	<i>190.100.159.255/26</i>
<i>Total =</i>	<i>$128 \times 64 = 8192$</i>	

Number of granted addresses to the ISP: 65,536

Number of allocated addresses by the ISP: 40,960

19.34 *Number of available addresses: 24,576*

Figure 19.9 *An example of address allocation and distribution by an ISP*



Exercise 21

An organization is granted the block 16.0.0.0/8. The administrator wants to create 500 fixed-length subnets.

- a. Find the subnet mask.*
- b. Find the number of addresses in each subnet.*
- c. Find the first and last addresses in subnet 1.*
- d. Find the first and last addresses in subnet 500 group*

Solution

a. *500 subnet = $2^9 = 512$. So, the total possible subnet is 512. Previous subnet mask is /8, now added /9 = $/(8+9)=/17$. The new subnet mask is /17*

b. *$2^{32-17} = 2^{15} = 32,768$ Addresses per subnet*

Exercise 21 (Continue)

- c. **Subnet 1:** The first address in this address is the beginning address of the block or **16.0.0.0**. To find the last address, we need to write 32,767 (one less than the number of addresses in each subnet) in base 256 (0.0.127.255) and add it to the first address (in base 256).

First address in subnet 1:	16	.	0	.	0	.	0
Number of addresses:	0	.	0	.	127	.	255
Last address in subnet 1:	16	.	0	.	127	.	255

d. **Subnet 500:**

Note that the subnet 500 is not the last possible subnet; it is the last subnet used by the organization. To find the first address in subnet 500, we need to add 16,351,232 (499×32678) in base 256 (0.249.128.0) to the first address in subnet 1. We have $16.0.0.0 + 0.249.128.0 = \mathbf{16.249.128.0}$. Now we can calculate the last address in subnet 500.

First address in subnet 500:	16	.	249	.	128	.	0
Number of addresses:	0	.	0	.	127	.	255
Last address in subnet 500:	16	.	249	.	255	.	255

Exercise 21 (Continue)

There are 512 subnets numbered from 0 to 511. Each second octet contains 2 subnets:

Subnet Number	Network Address
0	16.0.0.0/17
1	16.0.128.0/17
2	16.1.0.0/17
3	16.1.128.0/17
...	...
499	16.249.128.0/17

Table 19.3 *Addresses for private networks range different class IP*

<i>Range</i>			<i>Total</i>
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

19-2 IPv6 ADDRESSES

Despite all short-term solutions, address depletion is still a long-term problem for the Internet. This and other problems in the IP protocol itself have been the motivation for IPv6.

Topics discussed in this section:

Structure

Address Space



Note

An IPv6 address is 128 bits long.

Figure 19.14 *IPv6 address in binary and hexadecimal colon notation*

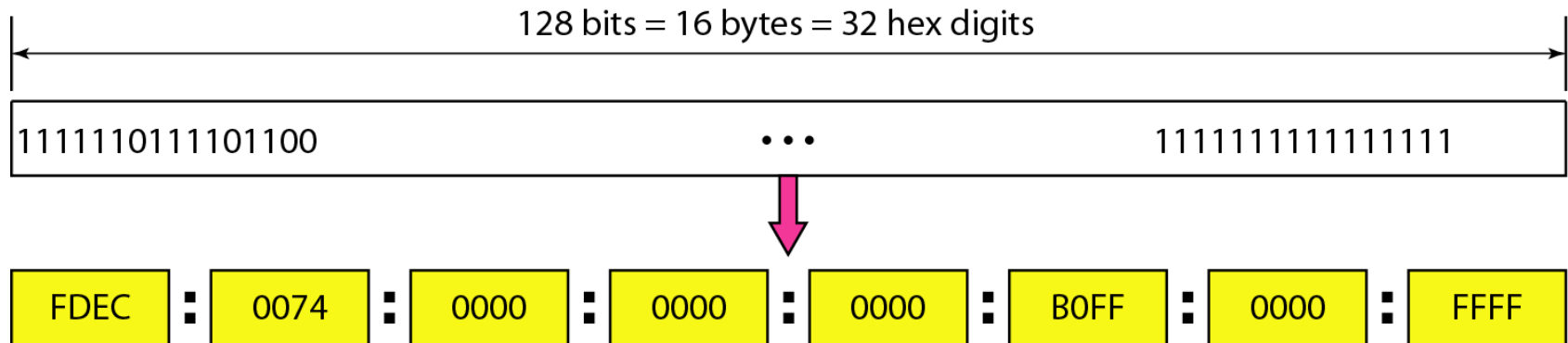
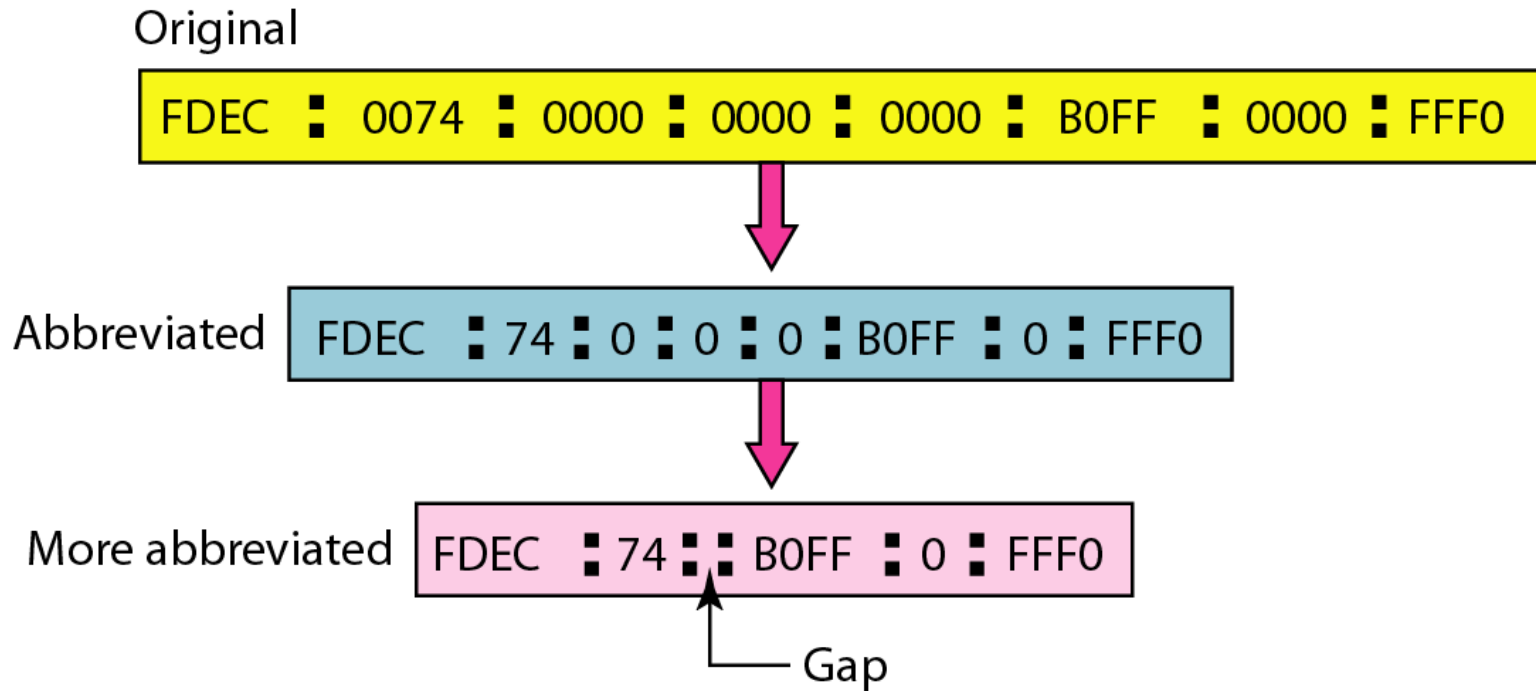


Figure 19.15 *Abbreviated IPv6 addresses*





Example 19.11

Expand the address 0:15::1:12:1213 to its original.

Solution

We first need to align the left side of the double colon to the left of the original pattern and the right side of the double colon to the right of the original pattern to find how many 0s we need to replace the double colon.

XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX
0: 15: : 1: 12:1213

This means that the original address is.

0000:0015:0000:0000:0000:0001:0012:1213

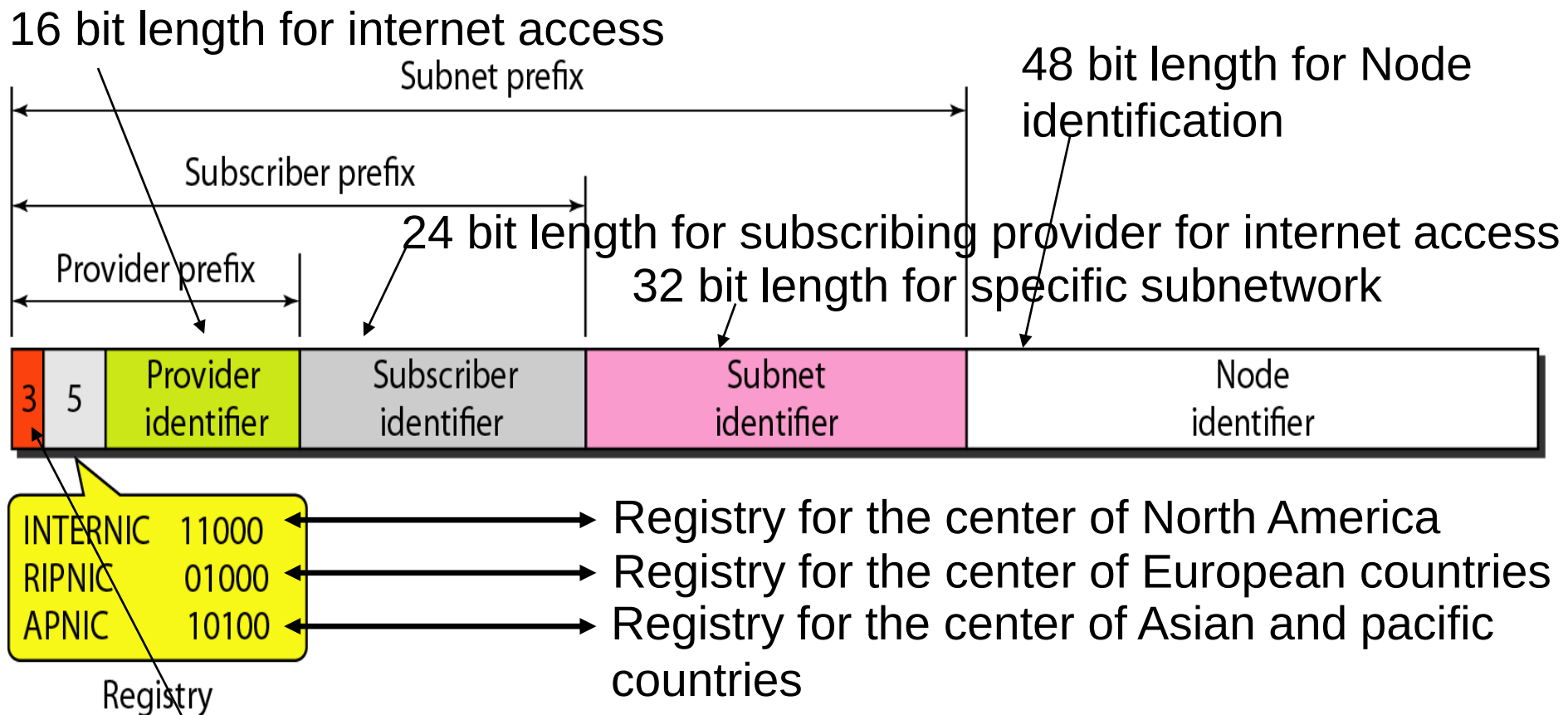
Table 19.5 *Type prefixes for IPv6 addresses*

<i>Type Prefix</i>	<i>Type</i>	<i>Fraction</i>
0000 0000	Reserved	1/256
0000 0001	Unassigned	1/256
0000 001	ISO network addresses	1/128
0000 010	IPX (Novell) network addresses	1/128
0000 011	Unassigned	1/128
0000 1	Unassigned	1/32
0001	Reserved	1/16
001	Reserved	1/8
010	Provider-based unicast addresses	1/8

Table 19.5 *Type prefixes for IPv6 addresses (continued)*

<i>Type Prefix</i>	<i>Type</i>	<i>Fraction</i>
011	Unassigned	1/8
100	Geographic-based unicast addresses	1/8
101	Unassigned	1/8
110	Unassigned	1/8
1110	Unassigned	1/16
1111 0	Unassigned	1/32
1111 10	Unassigned	1/64
1111 110	Unassigned	1/128
1111 1110 0	Unassigned	1/512
1111 1110 10	Link local addresses	1/1024
1111 1110 11	Site local addresses	1/1024
1111 1111	Multicast addresses	1/256

Figure 19.16 *Prefixes for provider-based unicast address*



3-bit defines the address as a provider based address

Figure 19.18 *Reserved addresses in IPv6*

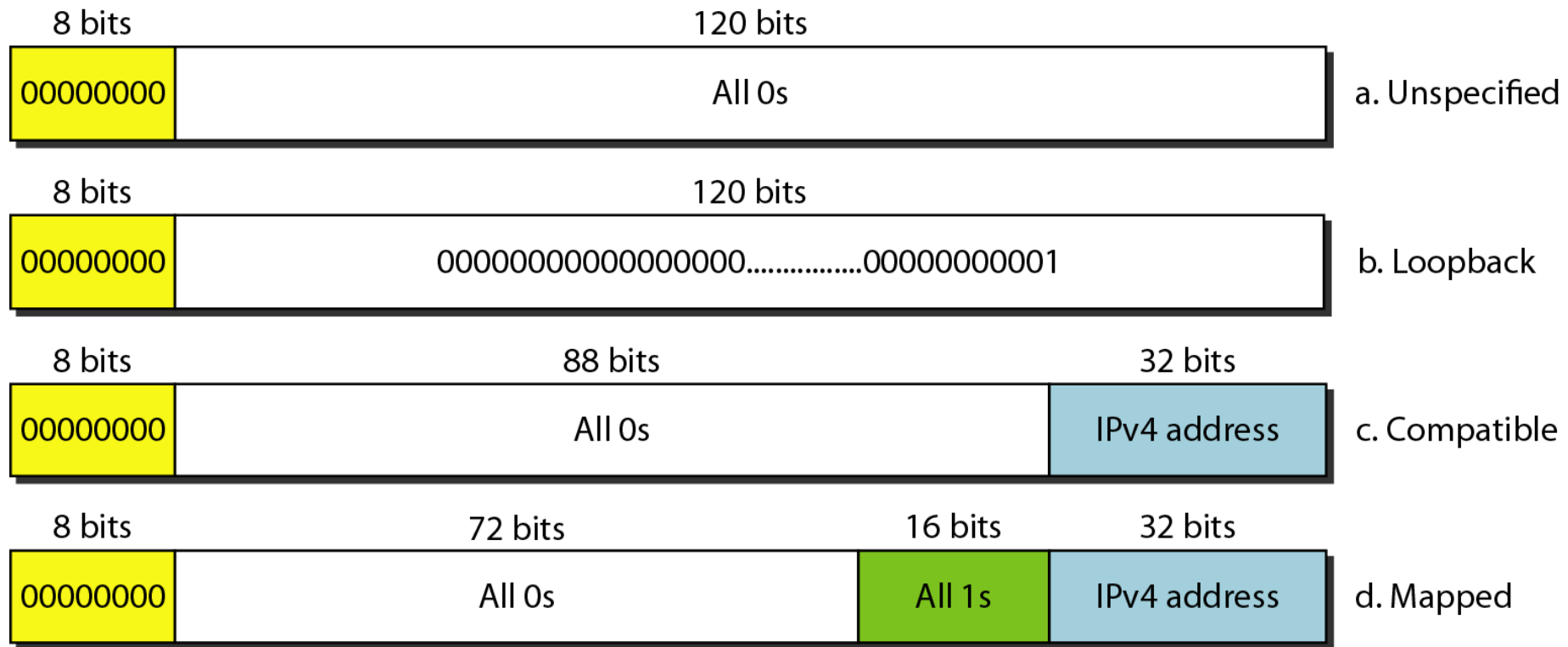


Figure 19.19 *Local addresses in IPv6*

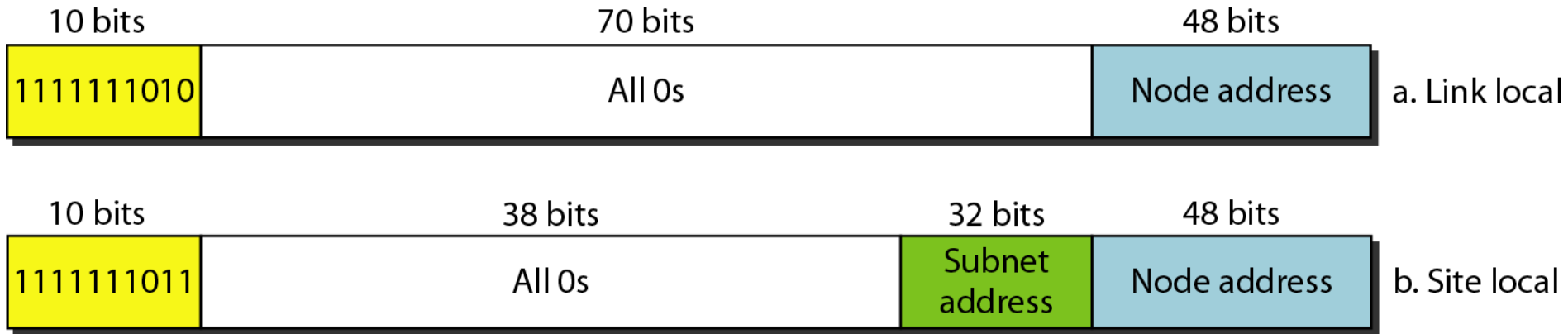


Figure 19.19 *Local addresses in IPv6*

Thanks